Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The value of $\lim_{x\to 0}\left(\frac{e^{x\log_e\left(2^x-1\right)}-\left(2^x-1\right)^x\sin x}{e^{x\log x}}\right)^{1/x}$ is equal to

(A) e

(B) $\left(\frac{1}{e}\right)\log 2$

(C) e $\log 2$

(D) None of these

Q.2

$\lim_{x\to 0}(\operatorname{cosec} x)^{1/\log x}$ equals

(A) 1

(B) -1

(C) e

(D) $1/\mathrm{e}$

Q.3

$\lim_{x\to\infty}\left(\frac{x^{100}}{e^x}+\left(\cos\frac{2}{x}\right)^{x^2}\right)=$

(A) e^{-1}

(B) e^{-4}

(C) $\left(1+e^{-2}\right)$

(D) e^{-2}

Q.4

The value of $\lim_{x\to\infty}\left(\frac{a_1^{1/x}+a_2^{1/x}+\ldots\ldots\ldots+a_n^{1/x}}{n}\right)^n$ $a_i>0, i=1,2,\ldots\ldots n$, is

(A) $a_1+a_2+\ldots\ldots\ldots\ldots+a_n$

(B) $e^{a_1+a_2+\ldots a_n}$

@IITJEE_Advanced

(C) $\frac{a_1+a_2+\ldots\ldots+a_n}{n}$

(D) $a_1 a_2 a_3 \ldots\ldots a_n$

Q.5

$\lim_{n\to\infty}\left(4^n+5^n\right)^{\frac{1}{n}}$ is equal to

(A) 4

(B) 5

(C) 5e

(D) $\frac{4}{5}e$

Q.6

If $a_1=1$ and $a_{n+1}=\left(4+3a_n\right)/\left(3+2a_n\right), n\geq 1$ and if $\lim_{(n\to\infty)} a_n=a$, then the value of a is

(A) $\sqrt{2}$

(B) $-\sqrt{2}$

(C) 2

(D) None of these

Q.7

Let $f(x)=\lim_{n\to\infty}\frac{\log(2+x)-x^{2n}\sin x}{1+x^{2n}}$. Then

(A) $\lim_{x\to 1^+} f(x)\neq\lim_{x\to 1^-} f(x)$

(B) $\lim_{x\to 1^+} f(x)=\sin 1$

(C) $\lim_{x\to 1^-} f(x)=$ does't exist

(D) None of these

Q.8

The value of $\lim_{x\to\infty}\left(\frac{3x-4}{3x+2}\right)^{\frac{x+1}{3}}$ is equal to :

(A) $e^{-1/3}$

(B) $e^{-2/3}$

(C) e^{-1}

(D) e^{-2}

Q.9

@IITJEE_Advanced

If $z_r = \cos\frac{r\alpha}{n^2} + i\sin\frac{r\alpha}{n^2}$, where $r = 1, 2, 3, \ldots, n$, then $\lim_{n\to\infty} z_1 z_2 z_3 \ldots z_n$ is equal to

(A) $\cos\alpha + i\sin\alpha$

(B) $\cos(\alpha/2) - i\sin(\alpha/2)$

(C) $e^{i\alpha/2}$

(D) $\sqrt[3]{e^{i\alpha}}$

Q.10

$\text{Lim}_{x\to\infty}\left(\frac{x^2+5x+3}{x^2+x+2}\right)^x$

(A) e^4

(B) e^2

(C) e^3

(D) 1

Q.11

If $\{x\}$ denotes the fractional part of x, then $\lim_{x\to[a]}\frac{e^{\{x\}}-\{x\}-1}{\{x\}^2}$, where [a] denotes the integral part of a, is equal to

(A) 0

(B) $\frac{1}{2}$

(C) e^{-2}

(D) None of these

Q.12

$\lim_{x\to 0}\left[\frac{\sin[x-3]}{[x-3]}\right]$, where $[.]$ denotes greatest integer function is

(A) 0

(B) 1

(C) does not exist

(D) $\sin 1$

Q.13

@IITJEE_Advanced

Let the sequence $< b_n >$ of real numbers satisfies the recurrence relation $b_{n+1} = \frac{1}{3}\left(2b_n + \frac{125}{b_n^2}\right), b_n \neq 0$. Then find $\lim_{n\to\infty} b_n$

(A) 10

(B) 15

(C) 5

(D) 25

Q.14

If $x_1 = 3$ and $x_{n+1} = \sqrt{2 + x_n}, n \geq 1$, then $\lim_{n\to\infty} x_n$ is equal to

(A) -1

(B) 2

(C) $\sqrt{5}$

(D) 3

Q.15

$\lim_{x\to 0} \frac{\sin x^4 - x^4 \cos x^4 + x^{20}}{x^4\left(e^{2x^4} - 1 - 2x^4\right)}$ is equal to

(A) 0

(B) $-1/6$

(C) $1/6$

(D) does not exist

Q.16

If $[\cdot]$ denotes the greatest integer function, then $\lim_{n\to\infty} \frac{[x]+[2x]+\ldots+[nx]}{n^2}$ is

(A) 0

(B) x

(C) $\frac{x}{2}$

(D) $\frac{x^2}{2}$

Q.17

The limit $\lim_{n\to\infty} \prod_{r=3}^{n} \frac{r^3-8}{r^3+8}$ is equal to

@IITJEE_Advanced

(A) $\frac{2}{7}$

(B) $\frac{1}{12}$

(C) $\frac{19}{52}$

(D) None of these

Q.18

The value of $\lim_{n \to \infty} \frac{1^2+1}{1-n^3} + \frac{2^2+2}{2-n^3} + \frac{3^2+3}{3-n^3} + \ ... \ ... \ + \frac{n^2+n}{n-n^3}$ is equal to

(A) $\frac{1}{2}$

(B) $\frac{-1}{2}$

(C) $\frac{1}{3}$

(D) $\frac{-1}{3}$

Q.19

$\lim_{x \to 1} \frac{(1-x)\left(1-x^2\right)\ldots\left(1-x^{2n}\right)}{\{(1-x)(1-x^2)\ldots(1-x^n)\}^2}, n \in N$, equals

(A) ${}^{2n}P_n$

(B) ${}^{2n}C_n$

(C) $(2n)!$

(D) None of these

Q.20

Let $f : R \to R$ be such that $f(1) = 3$ and $f'(1) = 6$.

Then $\lim_{x \to 0} \left(\frac{f(1+x)}{f(1)}\right)^{1/x}$ equals

(A) 1

(B) $e^{1/2}$

(C) e^2

(D) e^3

Answers & Solutions

Q.1 Answer:

$\left(\frac{1}{e}\right) \log 2$

Solution:

@IITJEE_Advanced

$\lim_{x\to 0}\left[\frac{(2^x-1)^x-(2^x-1)^x\sin x}{x^x}\right]^{1/x}$
$=\lim_{x\to 0}\frac{(2^x-1)}{x}\cdot[1-\sin x]^{1/x}$
$=\log 2\cdot\lim_{x\to 0}[1-\sin x]^{1/x}\left[\because \lim_{x\to 0}\frac{(2^x-1)}{x}=\log 2\right]$
which is 1^∞ form
$=(\log 2)\cdot\frac{1}{e}$

Q.2 Answer:

$1/\mathrm{e}$

Solution:

Let $A=\lim_{x\to 0}(\operatorname{cosec} x)^{1/\log x}$
$\Rightarrow \log A=\lim_{x\to 0}\frac{1}{\log x}\log\operatorname{cosec} x \quad =-\lim_{x\to 0}\frac{\log\sin x}{\log x} \quad =-\lim_{x\to 0}\frac{\cos x}{\sin x}x=-1$
$=A=1/e$

Q.3 Answer:

e^{-2}

Solution:

$\text{Consider } \lim_{x\to\infty}\left[\frac{x^{100}}{e^x}+\left(\cos\frac{2}{x}\right)^{x^2}\right]$
$=\lim_{x\to\infty}\frac{x^{100}}{e^x}+\lim_{x\to\infty}\left[\cos\left(\frac{2}{x}\right)\right]^{x^2}$
$=\lim_{x\to\infty}\frac{x^{100}}{e^x}=0 \text{ (Using L' Hopital's rule)}$
$\text{and } \lim_{x\to\infty}\left(\cos\frac{2}{x}\right)^{x^2} \text{ is of } (1^\infty) \text{ form}$
$=e^{\lim_{x\to\infty}x^2\left(\cos\frac{2}{x}-1\right)}$
$=e^{\lim_{t\to 0}t^{\frac{4}{2}}(\cos t-1)}\left(\text{Put } \frac{2}{x}=t\Rightarrow x=\frac{2}{t}\right)$
$=e^{-\lim_{t\to 0}\left(\frac{1-\cos t}{t^2}\right)\cdot 4}=e^{-\lim_{t\to 0}\left(\frac{\sin t}{2t}\right)4}=e^{-2}$

Q.4 Answer:

$a_1a_2a_3\ldots\ldots a_n$

Solution:

Putting $x=\frac{1}{y}$**, we get**

$\mathrm{L}=\textbf{limit}=\lim_{y\to 0}\left(\frac{a_1^y+a_2^y+\ldots\ldots+a_n^y}{n}\right)^{n/y}\ (\because x\to\infty, y\to 0)$
$\therefore \log_e \mathrm{L}=\lim_{y\to 0}\frac{n}{y}\cdot\log_e\frac{1}{n}\left(a_1^y+a_2^y+\ldots\ldots+a_n^y\right)\left(\frac{0}{0}\right)$
$=n\lim_{y\to 0}\frac{\left(\frac{a_1^y\log a_1+a_2^y\log a_2+\ldots\ldots+a_n^y\log a_n}{a_1^y+a_2^y+\ldots\ldots+a_n^y}\right)}{1}$ **[using L'Hospital rule]**
$=n\cdot\frac{\log(a_1a_2\ldots a_n)}{n}$
$\therefore \log \mathrm{L}=\log(a_1\cdot a_2\ldots a_n)\Rightarrow \mathrm{L}=a_1\cdot a_2\cdot a_3\ldots\ldots a_n$

@IITJEE_Advanced

Q.5 Answer:

5

Solution:

$\lim_{n\to\infty}\left(4^n+5^n\right)^{1/n}$
$=\lim_{n\to\infty}\left(5^n\right)^{1/n}\left[1+\left(\frac{4}{5}\right)^n\right]^{1/n}$
$=\lim_{n\to\infty}5\left[1+\left(\frac{4}{5}\right)^n\right]^{1/n}$
$=\lim_{n\to\infty}5(1+0)^{1/n}$
$=5\times1^0=5$

Q.6 Answer:

$\sqrt{2}$

Solution:

We have $a_{n+1}=\frac{4+3a_n}{3+2a_n}\Rightarrow\lim_{n\to\infty}a_{n+1}=\lim_{n\to\infty}\frac{4+3a_n}{3+2a_n}$
$\Rightarrow\quad a=\frac{4+3a}{3+2a}\Rightarrow 2a^2=4\Rightarrow a=\sqrt{2}$
$a\neq-\sqrt{2}$ **because each** $a_n>0$, **therefore** $\lim a_n=a>0$.

Q.7 Answer:

$\lim_{x\to1^+}f(x)\neq\lim_{x\to1^-}f(x)$

Solution:

For $|x|<1, x^{2n}\to0$ **as** $n\to\infty$ **and for** $|x|>1$,
$\frac{1}{x^{2n}}\to0$ **as** $n\to\infty$. **So**

$$f(x)=\begin{cases}\log(2+x) & \text{if } |x|<1\\ \lim_{n\to\infty}\frac{x^{-2n}\log(2+x)-\sin x}{x^{-2n}+1}=-\sin x & \text{if } |x|>1\\ \frac{1}{2}[\log(2+x)-\sin x] & \text{if } |x|=1\end{cases}$$

Thus $\lim_{x\to1+}f(x)=\lim_{x\to1}(-\sin x)=-\sin1$
and $\lim_{x\to1-}f(x)=\lim_{x\to1}\log(2+x)=\log3$

Q.8 Answer:

$e^{-2/3}$

Solution:

Consider $\lim_{x\to\infty}\left(\frac{3x-4}{3x+2}\right)^{\frac{x+1}{3}}=\lim_{x\to\infty}\left(\frac{3x+2-6}{3x+2}\right)^{\frac{x+1}{3}}$
$=\lim_{x\to\infty}\left(\frac{3x+2}{3x+2}-\frac{6}{3x+2}\right)^{\frac{x+1}{3}}=\lim_{x\to\infty}\left(1+\frac{-6}{3x+2}\right)^{\frac{x+1}{3}}$
$=\lim_{x\to\infty}\left(\left\{1+\frac{-6}{3x+2}\right\}^{\frac{3x+2}{-6}}\right)^{\frac{-6}{3x+2}\times\frac{x+1}{3}}$
$=\lim_{x\to\infty}[e]^{\frac{-6}{3x+2}\times\frac{x+1}{3}}\left(\because\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x=e\right)$
$=\lim_{x\to\infty}e^{\frac{-2x-2}{3x+2}}=\lim_{x\to\infty}e^{\frac{x(-2-2/x)}{x(3+2/x)}}=\lim_{x\to\infty}e^{\frac{-2-2/x}{3+2/x}}=e^{-2/3}$

Q.9 Answer:

@IITJEE_Advanced

$e^{i\alpha/2}$

Solution:

$z_r = \cos\frac{r\alpha}{n^2} + i\sin\frac{r\alpha}{n^2}$
$z_1 = \cos\frac{\alpha}{n^2} + i\sin\frac{\alpha}{n^2}$
$z_2 = \cos\frac{2\alpha}{n^2} + i\sin\frac{2\alpha}{n^2}; \ldots$
$\Rightarrow z_n = \cos\frac{n\alpha}{n^2} + i\sin\frac{n\alpha}{n^2}$
consider $\lim_{n\to\infty}(z_1 z_2 z_3 \ldots z_n)$
$= \lim_{n\to\infty}\left[\cos\left\{\frac{\alpha}{n^2}(1+2+3+\ldots.+n)\right\}\right.$
$\left.+i\sin\left\{\frac{\alpha}{n^2}(1+2+3+\ldots+n)\right\}\right]$
$= \lim_{n\to\infty}\left[\cos\frac{\alpha n(n+1)}{2n^2} + i\sin\frac{\alpha n(n+1)}{2n^2}\right]$
$= \lim_{x\to\infty}\frac{\cos\alpha(1+1/n)}{2} + \frac{i\sin\alpha(1+1/n)}{2}$
$\cos\frac{\alpha}{2} + i\sin\frac{\alpha}{2} = e^{\frac{i\alpha}{2}}$

Q.10 Answer:

e^4

Solution:

$\lim_{x\to\infty}\left(\frac{x^2+5x+3}{x^2+x+2}\right)^x$
$= \lim_{x\to\infty}\left(1+\frac{4x+1}{x^2+x+2}\right)^x$
$= \lim_{x\to\infty}\left[\left(1+\frac{4x+1}{x^2+x+2}\right)^{\frac{x^2+x+2}{4x+1}}\right]^{\frac{(4x+1)x}{x^2+x+2}}$
$= \lim_{x\to\infty}\frac{4x^2+x}{x^2+x+2} = e^{\lim_{x\to\infty}\frac{4+\frac{1}{x}}{1+\frac{1}{x}+\frac{2}{x^2}}} = e^4$

Q.11 Answer:

None of these

Solution:

Let $[a] = n$, **then** $\lim_{x\to n^-}\frac{e^{\{x\}}-\{x\}-1}{\{x\}^2}$
$= \lim_{\mathrm{h}\to 0}\frac{\mathrm{e}^{\{\mathrm{n}-\mathrm{h}\}}-\{\mathrm{n}-\mathrm{h}\}-1}{\{\mathrm{n}-\mathrm{h}\}^2} = \lim_{\mathrm{h}\to 0}\frac{\mathrm{e}^{1-\mathrm{h}}-(1-\mathrm{h})-1}{(1-\mathrm{h})^2} = \mathrm{e}-2$
and $\lim_{x\to n^+}\frac{e^{\{x\}}-\{x\}-1}{\{x\}^2} = \lim_{h\to 0}\frac{e^{\{n+h\}}-\{n+h\}-1}{\{n+h\}^2}$
$= \lim_{\mathrm{h}\to 0}\frac{\mathrm{e}^{\mathrm{h}}-\mathrm{h}-1}{\mathrm{h}^2} = \lim_{\mathrm{h}\to 0}\frac{1+\mathrm{h}+\frac{\mathrm{h}^2}{2!}+\frac{\mathrm{h}^3}{3!}+\ldots-\mathrm{h}-1}{\mathrm{h}^2} = \frac{1}{2}$
I imit does not exist.

Q.12 Answer:

does not exist

Solution:

$\lim_{x\to 0}\left[\frac{\sin[x-3]}{[x-3]}\right]$
For $x \to 0^+, [x-3] = -3$

@IITJEE_Advanced

$\therefore \frac{\sin[x-3]}{[x-3]} = \frac{\sin(-3)}{-3} = \frac{\sin 3}{3} \in (0,1)$
$\therefore \lim_{x\to 0^+} \frac{\sin[x-3]}{[x-3]} = 0$
For $x \to 0^-, [x-3] = -4$
$\therefore \frac{\sin[x-3]}{[x-3]} = \frac{\sin 4}{4}$ **lies in** $(-1,0)$
$\therefore \lim_{x\to 0^-} \left[\frac{\sin[x-3]}{[x-3]}\right] = -1$ $\quad \therefore$ **Limit does not exist.**

Q.13 Answer:

5

Solution:

Let $\lim_{n\to\infty} b_n = b$ **Now,**
$b_{n+1} = \frac{1}{3}\left(2b_n + \frac{125}{b_n^2}\right)$
$\lim_{n\to\infty} b_{n+1} = \frac{1}{3}\left(2\lim_{n\to\infty} b_n + \frac{125}{\lim_{n\to\infty} b_n^2}\right)$
$b = \frac{1}{3}\left(2b + \frac{125}{b^2}\right)$
$\Rightarrow \frac{b}{3} = \frac{125}{3b^2} \Rightarrow b^3 = 125$ **or** $b = 5.$

Q.14 Answer:

2

Solution:

$x_{n+1} = \sqrt{2+x_n} \Rightarrow \lim x_{n+1} = \sqrt{2+\lim x_n}$
$\Rightarrow t = \sqrt{2+t}$
$[\because \lim x_{n+1} = \lim x_n]$
$\Rightarrow t^2 - t - 2 = 0 \Rightarrow (t-2)(t+1) = 0 \Rightarrow t = 2$

Q.15 Answer:

$1/6$

Solution:

$$\lim_{x\to 0} \frac{\sin x^4 - x^4\cos x^4 + x^{20}}{x^4\left(e^{2x^4} - 1 - 2x^4\right)}$$
$$= \lim_{t\to 0} \frac{\sin t - t\cos t + t^5}{t\left(e^{2t} - 1 - 2t\right)}$$
$$= \lim_{t\to 0} \frac{t - \frac{t^3}{3!} + \frac{t^5}{5!} \ldots\ldots - t\left(1 - \frac{t^2}{2!} + \frac{t^4}{4!} \ldots\ldots\right) + t^5}{t\left(1 + 2t + \frac{4t^2}{2!} + \frac{8t^3}{3!} + \frac{16t^4}{4!} + \ldots\ldots - 1 - 2t\right)}$$
$$= \frac{-\frac{1}{6} + \frac{1}{2}}{2} = -\frac{-1+3}{12} = \frac{1}{6}$$

Q.16 Answer:

@IITJEE_Advanced

$\frac{x}{2}$

Solution:

$nx-1<[nx]\leq nx$. **Putting** $n=1,2,3,\ldots,n$

and adding them, $x\Sigma n-n<\Sigma[nx]\leq x\Sigma n$

$\therefore x\cdot\frac{\Sigma n}{n^2}-\frac{1}{n}<\frac{\Sigma[nx]}{n^2}\leq x\cdot\frac{\Sigma n}{n^2}$ **...(i)**

Now, $\lim_{n\to\infty}\left\{x\cdot\frac{\Sigma n}{n^2}-\frac{1}{n}\right\}=x\cdot\lim_{n\to\infty}\frac{\Sigma n}{n^2}-\lim_{n\to\infty}\frac{1}{n}=\frac{x}{2}$

As the two limits are equal, by (i) $\lim_{n\to\infty}\frac{\Sigma[nx]}{n^2}=\frac{x}{2}$

Q.17 Answer:

$\frac{2}{7}$

Solution:

$\mathrm{Lim}_{n\to\infty}\prod_{r=3}^{n}\frac{r^3-8}{r^3+8}$

$=\mathrm{Lim}_{n\to\infty}\left(\frac{3^3-8}{3^3+8}\right)\left(\frac{4^3-8}{4^3+8}\right)\ldots\ldots\left(\frac{n^3-8}{n^3+8}\right)$

$=\mathrm{Lim}_{n\to\infty}\left(\frac{3-2}{3+2}\cdot\frac{3^2+4+2(3)}{3^2+4-2(3)}\right)$

$\left(\frac{4-2}{4+2}\cdot\frac{4^2+4+2(4)}{4^2+4-2(4)}\right)\cdots\left(\frac{n-2}{n+2}\cdot\frac{n^2+4+2n}{n^2+4-2n}\right)$

$=\mathrm{Lim}_{\mathrm{n}\to\infty}\left(\frac{3-2}{3+2}\cdot\frac{4-2}{4+2}\cdot\frac{5-2}{5+2}\cdot\ldots\ldots\cdot\frac{\mathrm{n}-2}{\mathrm{n}+2}\right)\left(\frac{3^2+4+2(3)}{3^2+4-2(3)}\right.$

$\left.\frac{4^2+4+2(4)}{4^2+4-2(4)}\cdots\frac{\mathrm{n}^2+4+2\mathrm{n}}{\mathrm{n}^2+4-2\mathrm{n}}\right)$

$=\left(\frac{1.2.3.4.5.6.7.\ldots\ldots\ldots}{5.6.7.8.\ldots\ldots\ldots}\right)\left(\frac{19.28.39.52.63.\ldots\ldots}{7.12.19.28.39.52.\ldots\ldots}\right)$

$=\frac{1.2.3.4}{7.12}=\frac{2}{7}$

Q.18 Answer:

$\frac{-1}{3}$

Solution:

Clearly given function lies between

$\frac{\sum r^2+r}{1-n^3}$ **and** $\frac{\sum r^2+r}{n-n^3}$ **and**

$\lim_{n\to\infty}\frac{\sum r^2+r}{1-n^3}=\lim_{n\to\infty}\frac{\sum r^2+r}{n-n^3}=\frac{-1}{3}$.

Q.19 Answer:

${}^{2n}C_n$

Solution:

$\lim_{x\to1}\frac{(1-x)\left(1-x^2\right)\ldots\left(1-x^{2n}\right)}{\{(1-x)(1-x^2)\ldots(1-x^n)\}^2}$

$=\lim_{x\to1}\frac{\left(\frac{1-x}{1-x}\right)\left(\frac{1-x^2}{1-x}\right)\cdots\left(\frac{1-x^{2n}}{1-x}\right)}{\left(\left(\frac{1-x}{1-x}\right)\left(\frac{1-x^2}{1-x}\right)\cdots\left(\frac{1-x^n}{1-x}\right)\right)^2}$

$=\frac{1\times2\times3\ldots(2n)}{(1\times2\times3\ldots n)^2}=\frac{(2n)!}{n!n!}={}^{2n}C_n$

Q.20 Answer:

@IITJEE_Advanced

e^2

Solution:

Given that $f : R \to R$ **such that** $f(1) = 3$ **and** $f'(1) = 6$

Then $\lim_{x\to 0}\left[\frac{f(1+x)}{f(1)}\right]^{1/x}$

$= e^{\lim_{x\to 0}\frac{1}{x}[\log f(1+x) - \log f(1)]}$

$= e^{\lim_{x\to 0}\frac{\frac{1}{f(1+x)}f'(1+x)}{1}} = e^{\frac{f'(1)}{f(1)}} = e^{6/3} = e^2$

[Using L Hospital rule

@IITJEE_Advanced